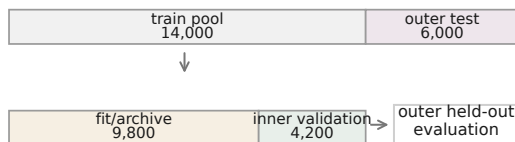


(a) trajectory-level split design

20,000 complete particle trajectories

splits are by particle trajectory, not by segment

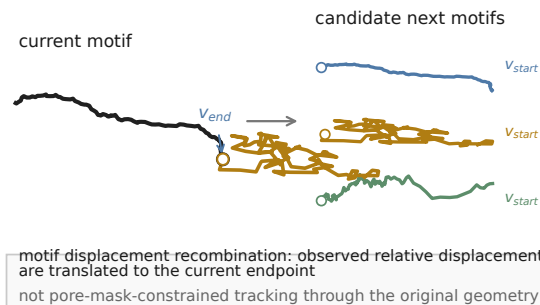


4 outer splits

3 inner validation repeats per outer split

archive motifs are cut only from fit trajectories

(b) motif recombination



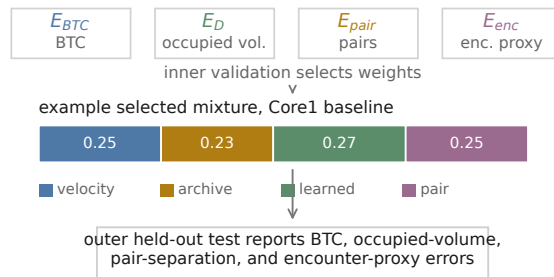
(c) candidate retained states



Each row defines what the next motif is allowed to know about the previous motif.

(d) inner validation objective

$$J = a_{BTC}E_{BTC} + a_DE_D + a_{pair}E_{pair} + a_{enc}E_{enc}$$



Selection and final evaluation use disjoint complete trajectories.